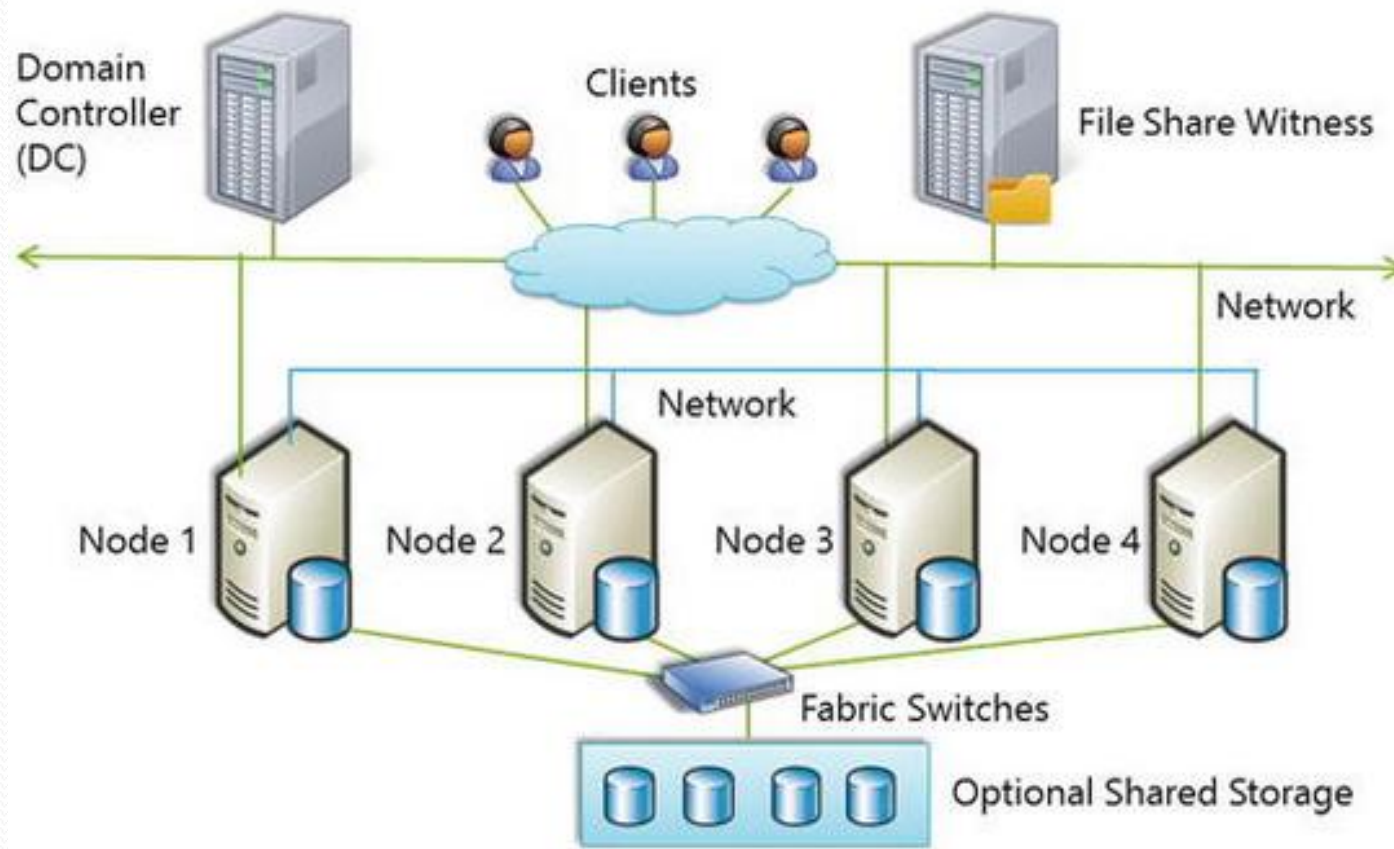




Failover Clustering



SOURCES: Configuring Advanced Windows Server® 2012 Services



Introduction

- High-availability clusters (also known as HA clusters or fail-over clusters) are groups of computers that support server applications that can be reliably utilized with a minimum amount of down-time. They operate by using high availability software to harness redundant computers in groups or clusters that provide continued service when system components fail. Without clustering, if a server running a particular application crashes, the application will be unavailable until the crashed server is fixed. HA clustering remedies this situation by detecting hardware/software faults, and immediately restarting the application on another system without requiring administrative intervention, a process known as failover. As part of this process, clustering software may configure the node before starting the application on it. For example, appropriate file systems may need to be imported and mounted, network hardware may have to be configured, and some supporting applications may need to be running as well.



Introduction

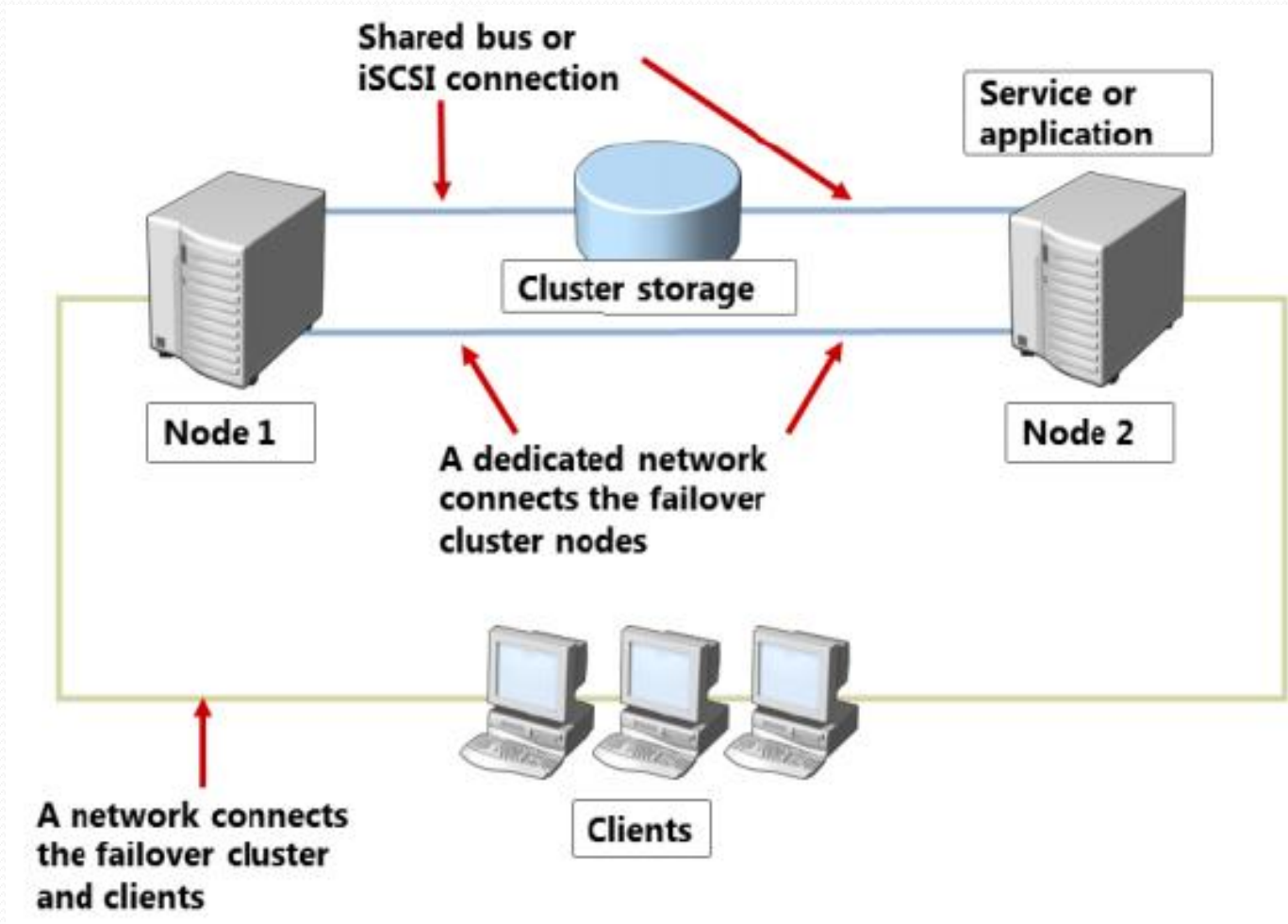
- HA clusters are often used for critical databases, file sharing on a network, business applications, and customer services such as electronic commerce websites.
- HA cluster implementations attempt to build redundancy into a cluster **to eliminate single points of failure**, including multiple network connections and data storage which is redundantly connected via storage area networks.



Failover Cluster Components

- As a failover cluster, a group of independent computers work together to increase the availability of applications and services. Physical cables and software connect the clustered servers. Servers that participate in the cluster are also known as nodes. If one of the cluster nodes fails, another node begins to provide services. This process is known as failover. With failover, users experience minimal to no service disruptions.

Failover Cluster Components





Failover Cluster Components

- A failover clustering solution consists of several components, which include:
 - ❑ **Nodes.** These are computers that are members of a failover cluster. These computers run Cluster service, and resources and applications associated with the cluster.
 - ❑ **Network.** This is a network across which cluster nodes can communicate with one another, and with clients. There are three types of networks that can be used in a cluster: public, private, and public-and-private.
 - ❑ **Resource.** This is an entity that is hosted by a node. It is managed by the Cluster service, and can be started, stopped, and moved to another node.



Failover Cluster Components

- ❑ **Cluster storage.** This is a storage system that is usually shared between cluster nodes.
- ❑ **Clients.** These are computers (or users) that are using the Cluster service.
- ❑ **Service or application.** This is a software entity that is presented to clients and used by clients.



Failover Cluster Components

- In a failover cluster, each node in the cluster:
 - ❑ Has full connectivity and communication with the other nodes in the cluster.
 - ❑ Is aware when another node joins or leaves the cluster. In addition, each node is aware when a node or resource is failing, and has the ability to take over those services.
 - ❑ Is connected to a network through which client computers can access the cluster.
 - ❑ Is usually connected through a shared bus or iSCSI connection to shared storage.
 - ❑ Is aware of the services or applications that are running locally, and the resources that are running on all other cluster nodes.

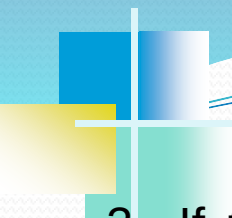


What Are Failover and Failback?

- Failover transfers from one node to another the responsibility for providing access to resources in a cluster. Failover can occur when an administrator intentionally moves resources to another node for maintenance, or due to unplanned downtime of a node due to hardware failure or other reasons. In addition, service failure on an active node can initiate failover to another node.
- A failover attempt consists of the following steps:

What Are Failover and Failback?

1. The Cluster service takes all the resources in the instance offline, in an order that is determined by the instance's dependency hierarchy. This means that dependent resources are taken offline first, followed by the resources on which they depend. For example, if an application depends on a physical disk resource, the Cluster service takes the application offline first, which enables the application to write changes to the disk before the disk is taken offline.
2. After all the resources are offline, the Cluster service attempts to transfer the instance to the node that is listed next on the instance's list of preferred owners.



What Are Failover and Failback?

3. If the Cluster service successfully moves the instance to another node, it attempts to bring all the resources back online. This time, it starts at the lowermost part of the dependency hierarchy. Failover is complete when all the resources are online on the new node.

What Are Failover and Failback?

- Once the offline node becomes active again, the Cluster service can fail back instances that were originally hosted on the offline node. When the Cluster service fails back an instance, it uses the same procedures that it performs during failover. The Cluster service takes all the resources in the instance offline, moves the instance, and then brings all the resources in the instance back online.



Failover Cluster Networks?

- Networks and network adapters are important parts of each cluster implementation. You cannot configure a cluster without configuring the networks that the cluster will use. A network can perform one of the following roles in a cluster:
 - **Private network.** A private network carries internal cluster communication. By using this network, cluster nodes exchange heartbeats and check for another node or nodes. The failover cluster authenticates all internal communication. However, administrators who are especially concerned about security may want to restrict internal communication to physically secure networks.

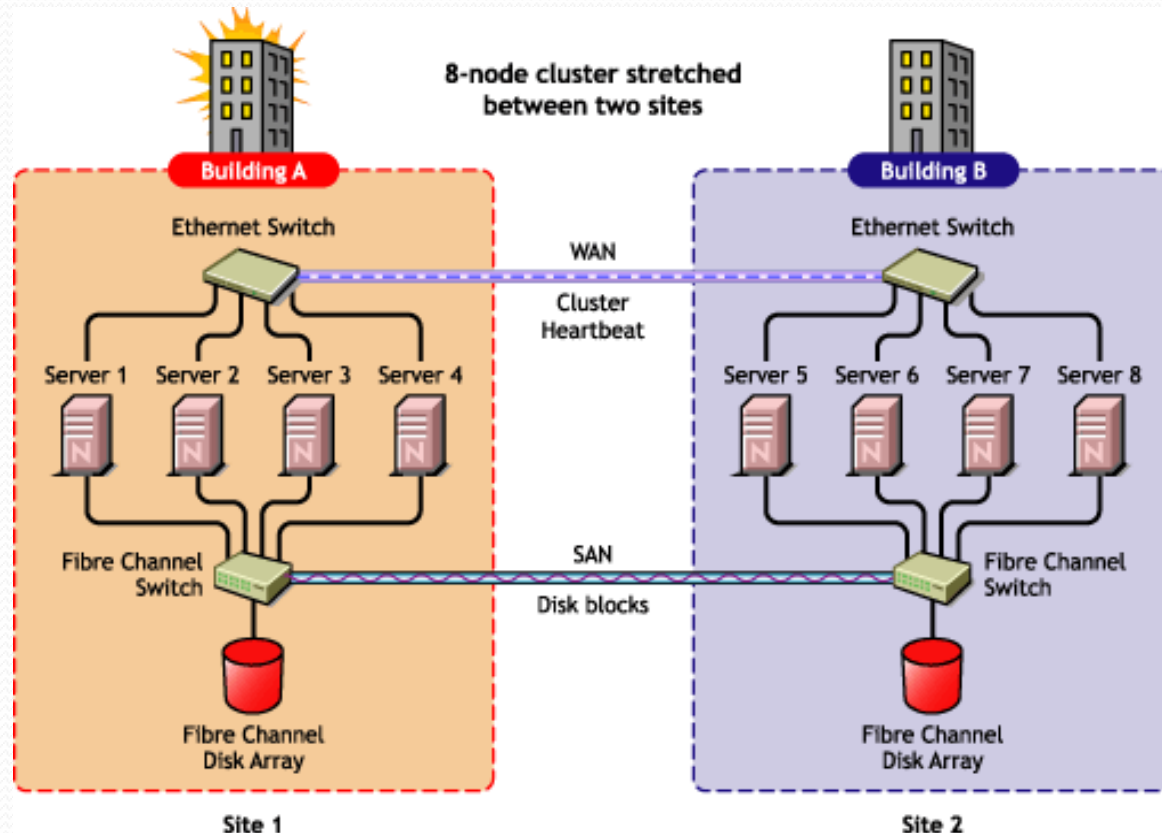


What Are Failover and Failback?

- **Public network.** A public network provides client systems with access to cluster application services. IP address resources are created on networks that provide clients with access to the Cluster service.
- **Public-and-private network.** A public-and-private network (also known as a mixed network) carries internal cluster communication and connects clients to cluster application services.

MetroCluster

- **MetroCluster** is an integrated high availability and disaster recovery solution for mission critical applications within a metropolitan and campus areas. • It combines software and hardware capabilities to provide disaster protection at a remote site providing no data loss, very fast recovery time, and automated recovery.



Load Balance vs Cluster

- Clustering (Backend): has a formal meaning. A cluster is a **group of resources** that are trying to achieve a common objective, and **are aware of one another**. Clustering usually involves setting up the resources (server usually) to exchange details on a particular channel (port) and keep exchanging their states, so **resource's is replicated at other places as well**. It usually also includes load balancing, wherein, the request is routed to one of the resources in cluster as per the load balancing policy.

Load Balance vs Cluster

- Load balancing (Frontend): can also happen **without clustering** when we have multiple independent servers that have the same setup, but other than that, are **unaware of each other**. Then, we can use a load balancer to forward requests to either one server or other, **but one server does not use the other server's resources**. Also, one resource does **not share its state with other resources**.

Load Balance vs Cluster

